

Cross-modal Affinity-aligned Multimodal Learning Analytics for Predicting Student Collaboration Satisfaction in Game-Based Learning



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What is Collaborative Game-based Learning?

Students work together in a game world to solve a real-world problem.



Collaborate

Students discuss and share ideas with teammates through chat.



Interact with NPCs

Students ask questions and receive information.



Collect Evidence

Students explore the island and collect clue items.



Make Inferences Together

Students organize evidence and build explanations on the whiteboard.



The game naturally captures rich behavioral data from multiple sources, reflecting how students collaborate.

What is Student Collaboration Satisfaction?

It reflects how students feel about quality of their collaboration experience.



Two Key Aspects



Group Listening

Are group members listening to one another?



Idea Building

Are students building on each other's ideas?

How is it measured?



Students answer an exit survey after each activity.

I am satisfied with how our group worked together.

1 2 3 4 5

Strongly Disagree Strongly Agree

4 Satisfaction Labels



High Satisfaction

high listening +
high building



Low Satisfaction

low listening +
low building



High Listening / Low Building

high listening +
low building



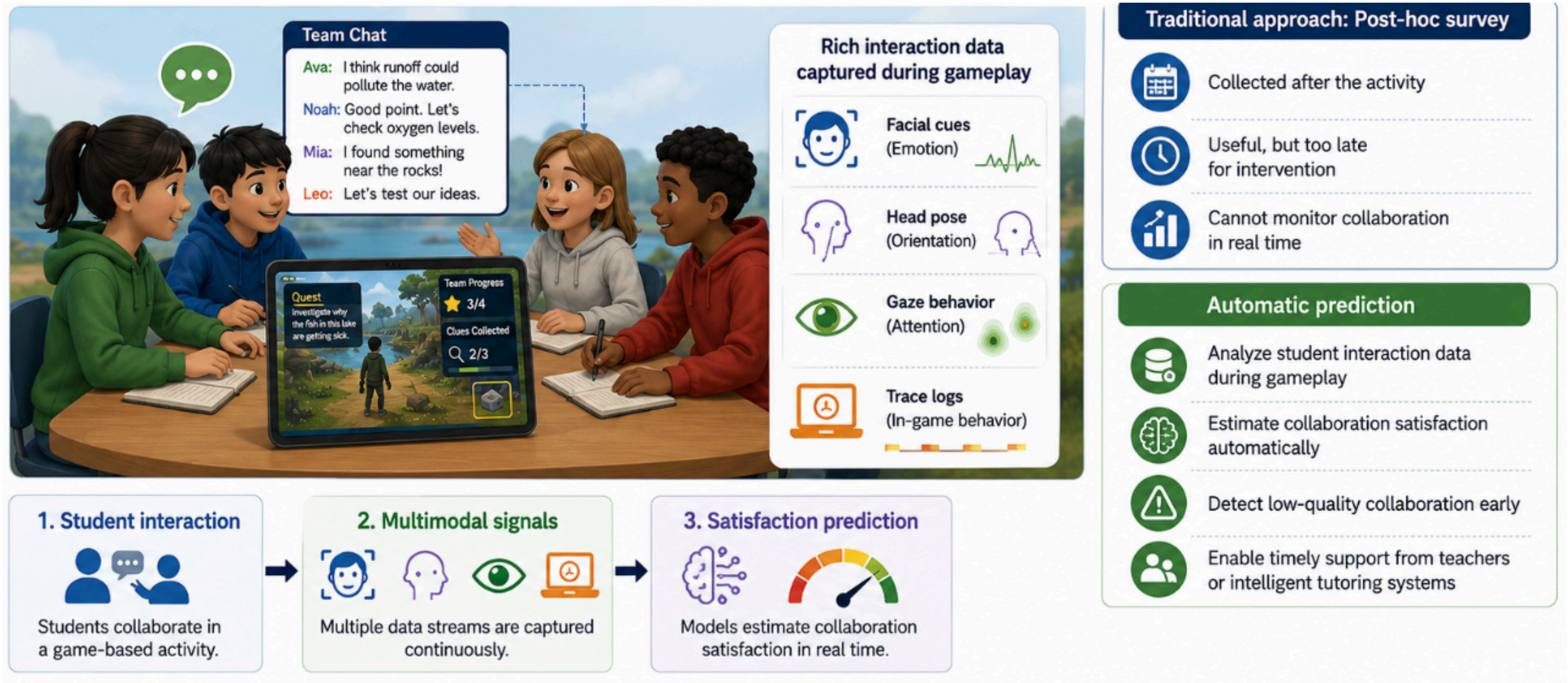
Low Listening / High Building

low listening +
high building

Goal: predict collaboration satisfaction automatically from student interaction observation.

Why Predict Collaboration Satisfaction Automatically?

Automatic prediction helps educators understand collaboration quality in time to provide support.



Goal: move from after-class evaluation to timely, data-driven support for better collaboration.

What is Multimodal Learning Analytics?

Combine multiple student data streams to understand learning & collaboration more completely.

4 Modalities

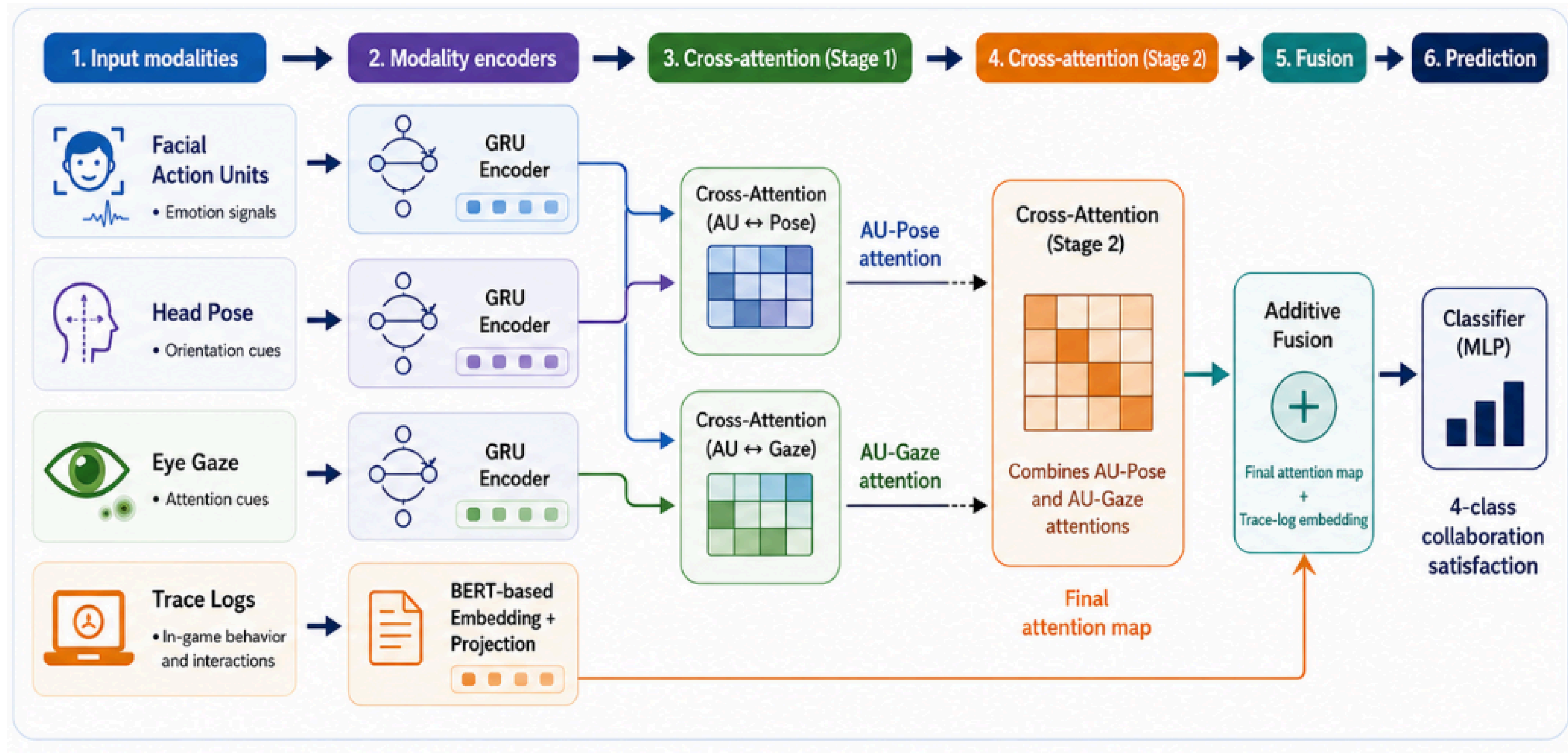
- Facial AU — emotion (smiling, frowning, tension)
- Head Pose — orientation / attention (nodding, leaning)
- Eye Gaze — visual attention (where students look)
- Trace Logs — in-game behavior (chat, NPC interaction, evidence)

Why multimodal?

- Each modality captures a different behavioral dimension
- No single signal tells the full story
- Combined → richer, more accurate student modeling
- Pipeline: Multi-stream capture → Align & fuse → Collaboration insights
- Key idea: Complementary signals > any single modality alone.

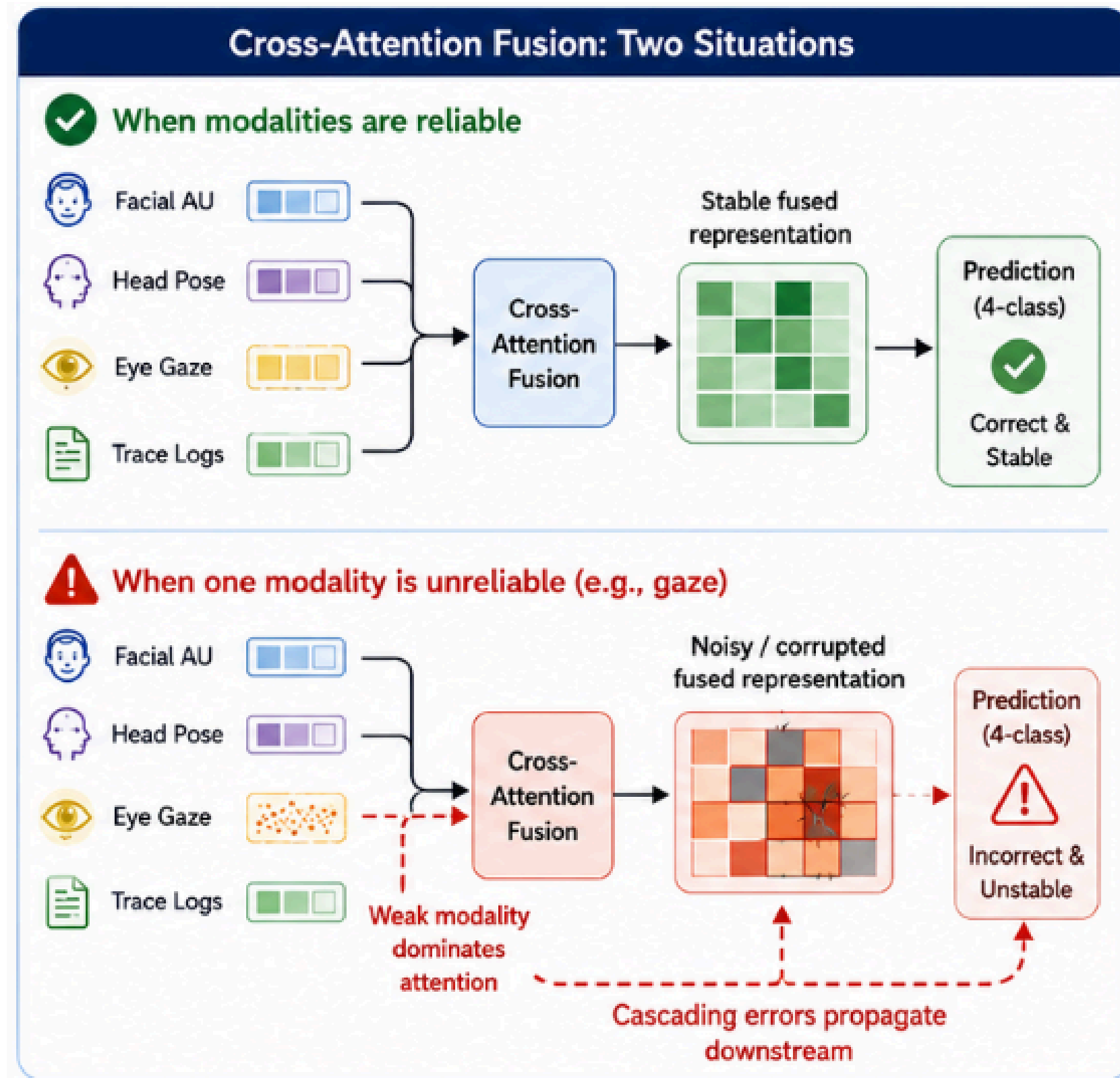
Prior Work: Multimodal Cross-Attention Model

A previous EDM 2024 approach predicts collaboration satisfaction by fusing multiple modalities with cross-attention.



Limitation: Multimodal Fusion Can Be Brittle

When one modality degrades, cross-attention lets it dominate → cascading errors through the whole pipeline.



The failure mode

- Gaze: noisy / missing / inconsistent across students
- Cross-attention has no mechanism to down-weight weak signals
- One bad modality corrupts the fused representation → wrong prediction

[4] UMCL: Unimodal-generated Multimodal Contrastive Learning for Cross-compression-rate Deepfake Detection, IJCV 2026

Limitation: Multimodal Fusion Can Be Brittle

Some modalities are noisy or inconsistent, and implicit fusion can let them harm the whole prediction pipeline.

Why it matters

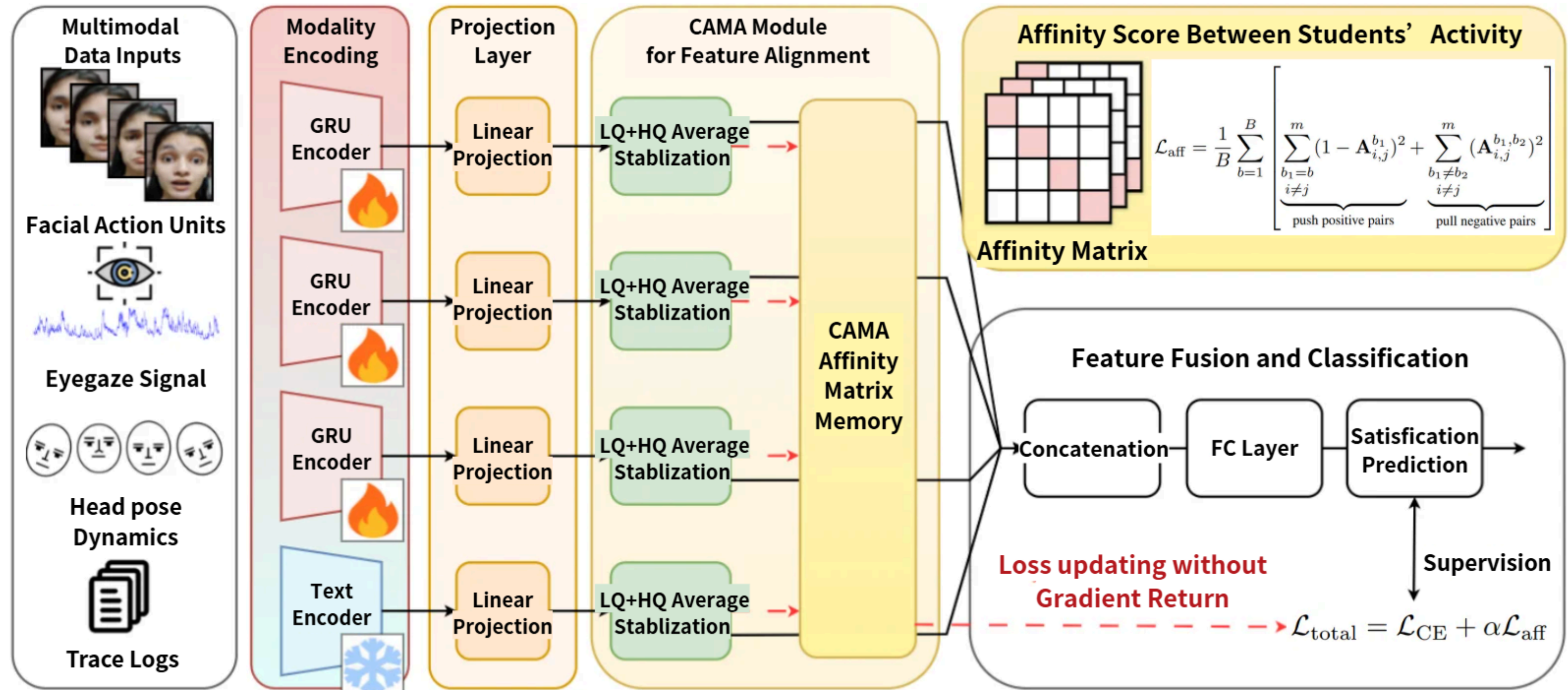
- Modality quality varies across students and sessions
- Implicit attention may over-trust weak signals
- Discarding the modality entirely also loses useful signal

Motivation

- Need explicit alignment that suppresses unreliable modalities — without throwing them away
- Takeaway: Robust multimodal modeling must handle modality degradation gracefully.

Our Approach: AAMLA with CAMA

Affinity-Alignment Multimodal Learning Analytics for robust collaboration satisfaction prediction.



AAMLA: Robust Multimodal Learning Through Modality Alignment with CAMA

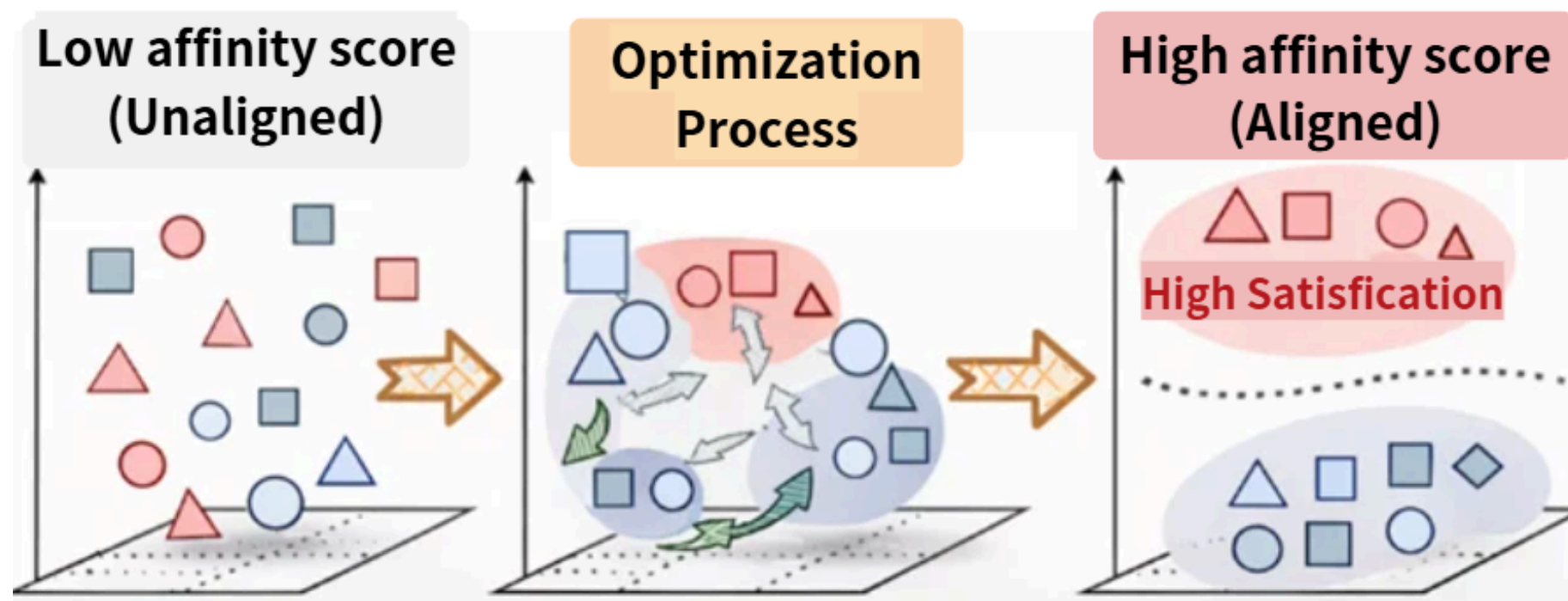
CAMA explicitly aligns modalities in a shared semantic space — reducing the impact of unreliable ones.

LQ+HQ Average Stabilization

$$\begin{aligned}\hat{\mathbf{E}}_{\text{AU}} &= \frac{1}{2}(\mathbf{E}_{\text{AU}}^{\text{orig}} + \mathbf{E}_{\text{AU}}^{\text{deg}}), & \hat{\mathbf{E}}_{\text{pose}} &= \frac{1}{2}(\mathbf{E}_{\text{pose}}^{\text{orig}} + \mathbf{E}_{\text{pose}}^{\text{deg}}), \\ \hat{\mathbf{E}}_{\text{gaze}} &= \frac{1}{2}(\mathbf{E}_{\text{gaze}}^{\text{orig}} + \mathbf{E}_{\text{gaze}}^{\text{deg}}), & \hat{\mathbf{E}}_{\text{trace}} &= \frac{1}{2}(\mathbf{E}_{\text{trace}}^{\text{orig}} + \mathbf{E}_{\text{trace}}^{\text{deg}}),\end{aligned}$$

- Feed each sample twice: clean + degraded
- Average before entering CAMA → stable batch-level representation under degradation
- Prevents a single noisy signal from corrupting the affinity matrix

Cross-modal Affinity-guided Modality Alignment (CAMA)



- Before: modality embeddings scattered, no semantic coherence
- CAMA optimization pulls same-class modalities together, pushes different-class apart
- After: compact, class-separable clusters even under degradation

AAMLA: Robust Multimodal Learning Through Modality Alignment with CAMA

CAMA explicitly aligns modalities in a shared semantic space — reducing the impact of unreliable ones.

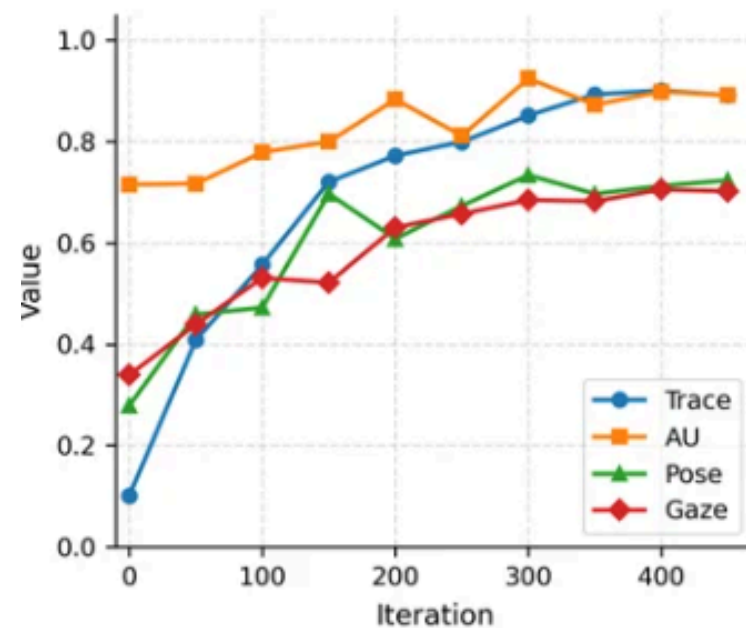
Affinity Loss

$$\mathcal{L}_{\text{aff}} = \frac{1}{B} \sum_{b=1}^B \left[\underbrace{\sum_{\substack{b_1=b \\ i \neq j}}^m (1 - \mathbf{A}_{i,j}^{b_1})^2}_{\text{push positive pairs}} + \underbrace{\sum_{\substack{b_1 \neq b_2 \\ i \neq j}}^m (\mathbf{A}_{i,j}^{b_1, b_2})^2}_{\text{pull negative pairs}} \right]$$

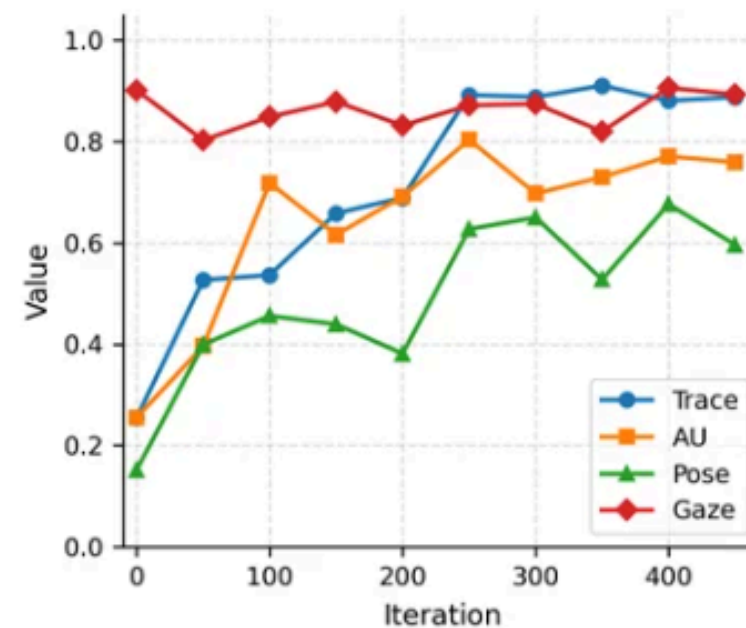
- Positive pairs $(1 - A)^2$ — same student activity, different modalities → force affinity toward 1 (align them)
- Negative pairs $(A)^2$ — different student activities → force affinity toward 0 (separate them)

Similar multimodal features are pulled together. Noisy ones can't align → down-weighted automatically.

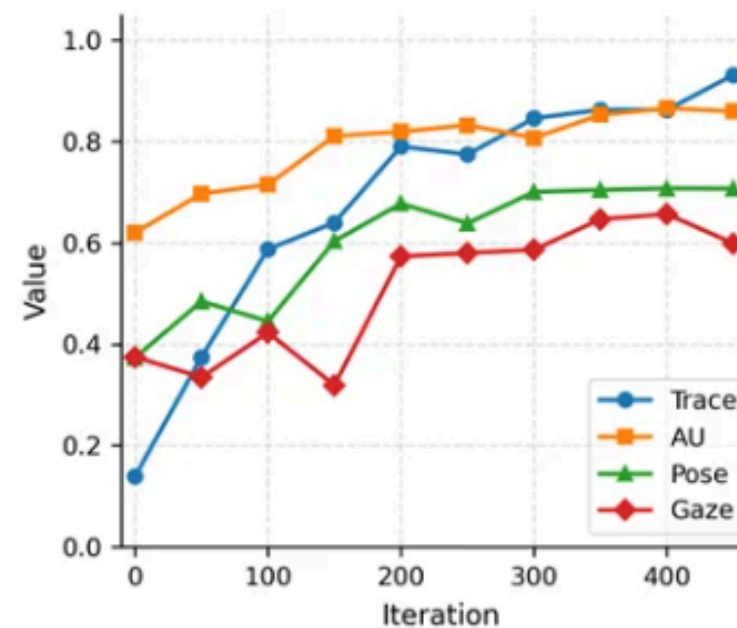
Results



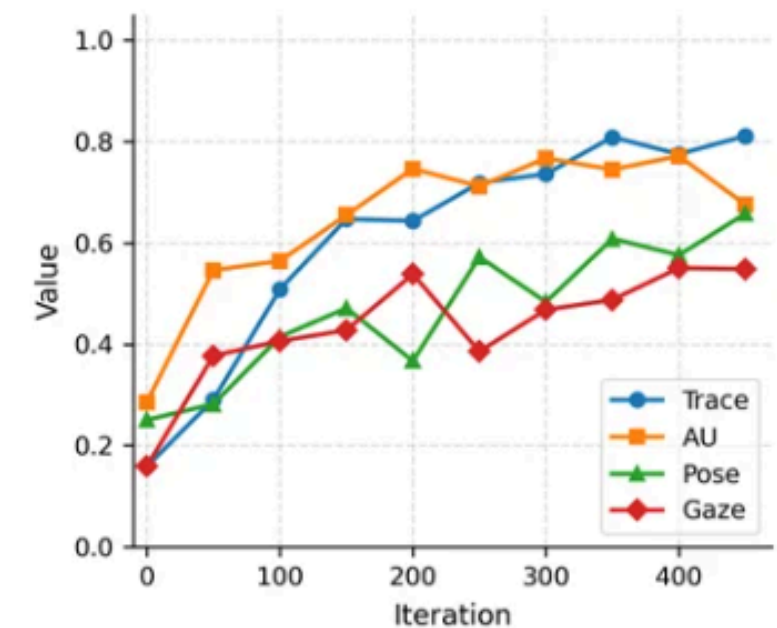
(a) High-satisfaction affinity scores



(b) Low-satisfaction affinity scores

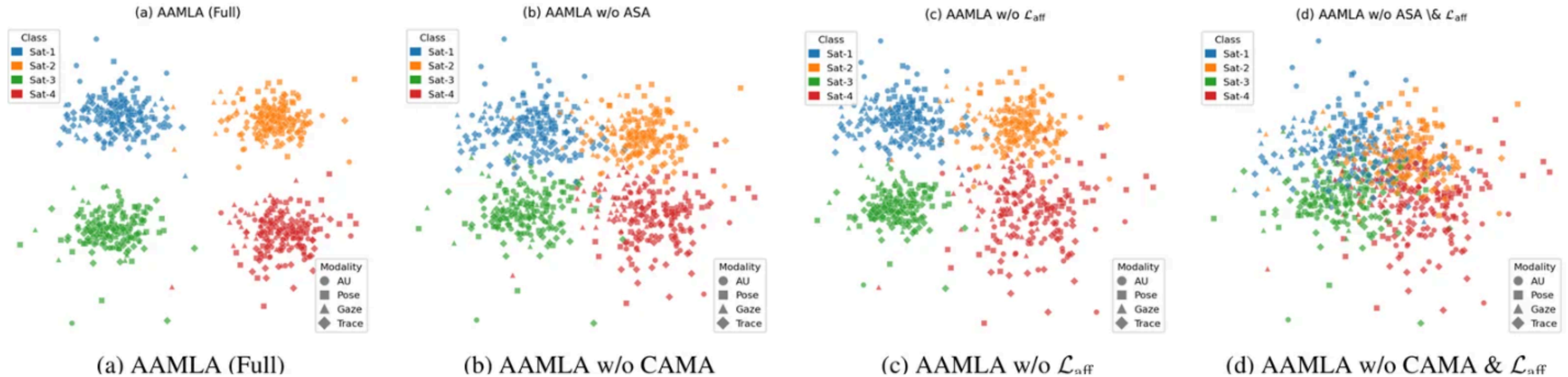


(c) Degraded high-satisfaction affinity scores



(d) Degraded low-satisfaction affinity scores

t-SNE Ablation: What Each Component Contributes



What this proves

- CAMA alone $\rightarrow$ inter-modal coherence
- $\mathcal{L}_{aff}$ alone $\rightarrow$ sharper class boundaries
- Both together $\rightarrow$ compact and discriminative

CAMA structures the feature space via affinity matrices. **Affinity Loss** enforces semantic consistency through contrastive supervision.

Each plays a distinct role; neither alone is sufficient. Together they produce representations that are both coherent and class-separable.

AAMLA Results: State-of-the-Art Performance and Strong Robustness

Results: AAMLA vs. Baselines

Model	F1-Score	Accuracy
AU (Unimodal)	0.66 $\pm$ 0.03	0.66 $\pm$ 0.03
Pose (Unimodal)	0.66 $\pm$ 0.04	0.66 $\pm$ 0.04
Gaze (Unimodal)	0.65 $\pm$ 0.05	0.65 $\pm$ 0.05
Trace (Unimodal)	0.65 $\pm$ 0.04	0.65 $\pm$ 0.04
Cross-Attention [1]	0.72 $\pm$ 0.03	0.72 $\pm$ 0.03
AAMLA (Ours)	0.79 $\pm$0.02	0.77 $\pm$0.02

Robustness under degradation (F1)

Modality	Degradation	Cross-Attn [1]	AAMLA (Ours)
Gaze	Dropout 30%	0.68	0.77
	Dropout 50%	0.61	0.75
	Dropout 70%	0.52	0.72
AU + Pose	$\mathcal{N}(0, 0.01)$	0.65	0.78
	$\mathcal{N}(0, 0.05)$	0.54	0.72
Full Absence	w/o AU	0.54	0.73
	w/o Pose	0.53	0.72
	w/o Gaze	0.52	0.68
	w/o Trace	0.56	0.70

- AAMLA outperforms Cross-Attention by +9.7% F1 under clean conditions
- Under severe degradation, Cross-Attention collapses; AAMLA stays robust

Explicit affinity alignment doesn't just improve accuracy — it preserves it when things go wrong.

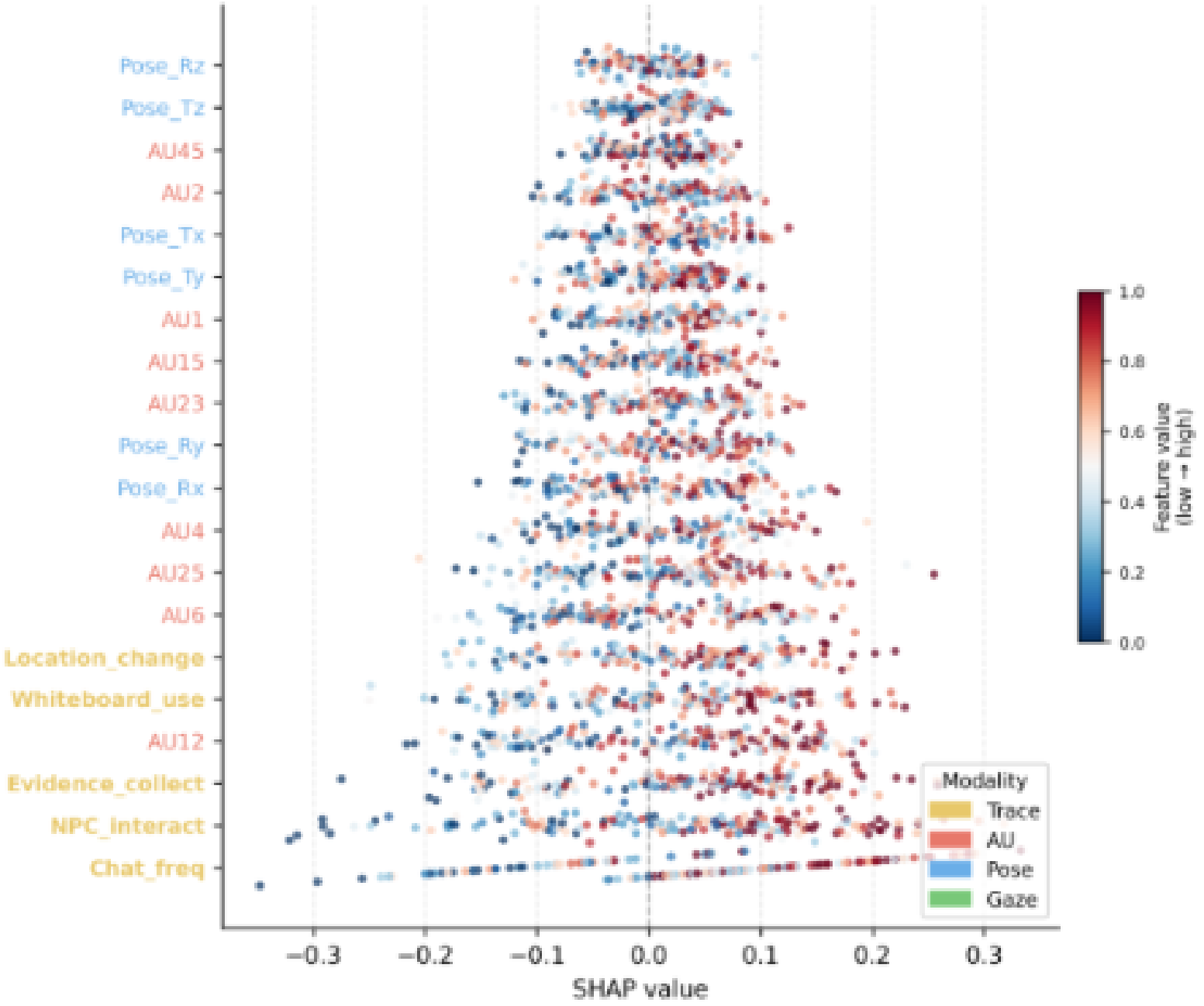
Modality Combination & Feature Importance Analysis

Which combinations work best (AAMLA)

SHAP: What actually drives prediction

Modality Combination	Cross-Attn [1]		AAMLA (Ours)	
	F1	Acc	F1	Acc
Full Multimodal	0.72	0.72	0.79	0.77
Trace + AU	0.70	0.70	0.73	0.74
Trace + Pose	0.69	0.69	0.72	0.74
Trace + Gaze	0.65	0.65	0.70	0.71
AU + Pose	0.68	0.67	0.71	0.72
AU + Gaze	0.65	0.65	0.70	0.71
Pose + Gaze	0.64	0.64	0.69	0.68
Trace + AU + Pose	0.70	0.70	0.74	0.76

- Trace logs are the backbone of collaboration signal.
- CAMA's job is to protect that signal from noisy modalities corrupting the fusion.



Conclusion

- **What we built AAMLA** — AAMLA predicts student collaboration satisfaction in real classroom deployments — where sensors fail, gaze is unreliable, and data is messy by design.
- **What makes it different** — Rather than assuming every modality is trustworthy, CAMA explicitly models inter-modal relationships via affinity matrices and learns to suppress noisy signals adaptively.
- **Broader implication** — Explicit multimodal alignment is a practical necessity in educational settings, where modality reliability varies across students, sessions, and hardware conditions.



**Thanks
Your
Attention!**

**Cross-modal Affinity-aligned Multimodal Learning
Analytics for Predicting Student Collaboration
Satisfaction in Game-Based Learning**